

Diag Solutions

1. 1. In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PRQ$. If $PS = 2$ cm, $SQ = 3$ cm, and $PT = 2.5$ cm, find TR .

Answer:

1.5 cm

Solution:

1. Show $\triangle PST \sim \triangle PRQ$ by AA similarity criterion.
2. Write the ratio of sides and solve for TR .

Derivation:

$$\begin{aligned} \triangle PST &\sim \triangle PRQ \\ \Rightarrow \frac{PS}{PR} = \frac{PT}{PQ} \text{ is incorrect; use } \frac{PS}{PR} = \frac{PT}{PQ} \text{ is not needed, use } \frac{PS}{PT} = \frac{PR}{PQ} \text{ is not right; use } \frac{PS}{PR} = \frac{PT}{PQ} \\ &\Rightarrow \frac{PS}{PT + TR} = \frac{PT}{PS + SQ} \\ &\Rightarrow \frac{2}{2.5 + TR} = \frac{2.5}{5} \\ &\Rightarrow 10 = 6.25 + 2.5TR \\ &\Rightarrow 3.75 = 2.5TR \\ &\Rightarrow TR = 1.5 \text{ cm} \end{aligned}$$

Triangles — Similarity of Triangles

2. 2. Diagonals of a trapezium ABCD with $AB \parallel DC$ intersect each other at the point O . If $AB = 2CD$, find the ratio of the areas of triangles $\triangle AOB$ and $\triangle COD$.

Answer:

4 : 1

Solution:

1. Prove $\triangle AOB \sim \triangle COD$ by AA similarity.
2. Use the ratio of areas theorem.

Derivation:

$$\begin{aligned} \triangle AOB &\sim \triangle COD \\ \Rightarrow \frac{\text{Area}(AOB)}{\text{Area}(COD)} &= \left(\frac{AB}{CD}\right)^2 = \left(\frac{2CD}{CD}\right)^2 = 2^2 = 4 \end{aligned}$$

Triangles — Areas of Similar Triangles

3. 3. In $\triangle ABC$, $DE \parallel BC$ such that $AD = (4x - 3)$ cm, $AE = (8x - 7)$ cm, $BD = (3x - 1)$ cm, and $EC = (5x - 3)$ cm. Find the value of x .

Answer:

$$x = 1$$

Solution:

1. Since $DE \parallel BC$, by Basic Proportionality Theorem (BPT), $\frac{AD}{DB} = \frac{AE}{EC}$.
2. Substitute the given values into the equation: $\frac{4x-3}{3x-1} = \frac{8x-7}{5x-3}$.
3. Cross-multiply to get: $(4x - 3)(5x - 3) = (8x - 7)(3x - 1)$.
4. Solve the resulting linear equation to find x .

Derivation:

$$\begin{aligned} 20x^2 - 12x - 15x + 9 &= 24x^2 - 8x - 21x + 7 \\ \implies 20x^2 - 27x + 9 &= 24x^2 - 29x + 7 \\ \implies 4x^2 - 2x - 2 &= 0 \\ \implies 2x^2 - x - 1 &= 0 \\ \implies (2x + 1)(x - 1) &= 0. \text{ Since } x > 1, x = 1. \end{aligned}$$

Triangles — Basic Proportionality Theorem

4. 4. Two poles of heights 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

Answer:

$$13m$$

Solution:

1. Let $AB = 6$ m and $CD = 11$ m be the two poles. The distance $BD = 12$ m.
2. Draw a line from A parallel to BD meeting CD at E . Then $CE = CD - ED = 11 - 6 = 5$ m.
3. In right-angled triangle $\triangle AEC$, use the Pythagorean theorem to find the hypotenuse AC .
4. $AC^2 = AE^2 + CE^2$.

Derivation:

$$AE = BD = 12 \text{ m. } CE = 11 - 6 = 5 \text{ m. } AC = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13 \text{ m.}$$

Triangles — Pythagoras Theorem

5. 5. In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PRQ$. Show that $\triangle PQR \sim \triangle PTS$.

Answer:

Proof

Solution:

1. Identify the common angle in both triangles: $\angle QPR = \angle TPS$ (Common).
2. Use the given condition $\angle PST = \angle PRQ$.
3. Conclude similarity by AA (Angle-Angle) criterion.

Derivation:

In $\triangle PQR$ and $\triangle PTS$: $\angle P = \angle P$ (Common), $\angle PST = \angle PRQ$ (Given). Therefore, $\triangle PQR \sim \triangle PTS$ by AA similarity criterion.

Triangles — Criteria for Similarity of Triangles

6. 6. The areas of two similar triangles are 121 cm^2 and 64 cm^2 respectively. If the median of the first triangle is 12.1 cm , find the corresponding median of the other triangle.

Answer:

8.8 cm

Solution:

1. Use the theorem: Ratio of areas of similar triangles is equal to the ratio of the squares of their corresponding medians.
2. Set up the equation: $\frac{\text{Area}_1}{\text{Area}_2} = \left(\frac{m_1}{m_2}\right)^2$.
3. Substitute values: $\frac{121}{64} = \left(\frac{12.1}{m_2}\right)^2$.
4. Take the square root and solve for m_2 .

Derivation:

$$\begin{aligned} \frac{11}{8} &= \frac{12.1}{m_2} \\ \Rightarrow m_2 &= \frac{12.1 \times 8}{11} = 1.1 \times 8 = 8.8 \text{ cm.} \end{aligned}$$

Triangles — Areas of Similar Triangles

7. 7. In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\angle BAC = 90^\circ$.

Answer:

Proof

Solution:

1. Since $AD \perp BC$, we have two right triangles $\triangle ADB$ and $\triangle ADC$.
2. Express the sides using Pythagoras: $AB^2 = AD^2 + BD^2$ and $AC^2 = AD^2 + CD^2$.

3. Add them and use the given condition $AD^2 = BD \cdot CD$ to show $AB^2 + AC^2 = BC^2$.

Derivation:

$AB^2 + AC^2 = AD^2 + BD^2 + AD^2 + CD^2 = 2AD^2 + BD^2 + CD^2$. Substitute $AD^2 = BD \cdot CD$: $AB^2 + AC^2 = 2(BD \cdot CD) + BD^2 + CD^2 = (BD + CD)^2 = BC^2$. By converse of Pythagoras, $\angle BAC = 90^\circ$.

Triangles — Pythagoras Theorem

8. 8. State and prove the Basic Proportionality Theorem (Thales Theorem).

Answer:

If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

Solution:

1. Draw a triangle ABC with a line DE parallel to BC intersecting AB at D and AC at E.
2. Construct perpendiculars DM on AC and EN on AB. Join BE and CD.
3. Calculate the area of triangles ADE, BDE, and CDE.
4. Use the property that triangles on the same base and between the same parallels have equal areas.
5. Divide the ratio of areas to prove the theorem.

Derivation: In $\triangle ADE$ and $\triangle BDE$, $\frac{ar(ADE)}{ar(BDE)} = \frac{\frac{1}{2} \cdot AD \cdot EN}{\frac{1}{2} \cdot BD \cdot EN} = \frac{AD}{BD}$. Similarly, $\frac{ar(ADE)}{ar(CDE)} = \frac{AE}{CE}$. Since $ar(BDE) = ar(CDE)$ (same base DE and same parallels DE $\parallel$ BC), it follows that $\frac{AD}{BD} = \frac{AE}{CE}$. *Triangles — Basic Proportionality Theorem*

9. 9. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

Answer:

$$\frac{ar(\triangle ABC)}{ar(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$$

Solution:

1. Draw two similar triangles ABC and PQR.
2. Draw altitudes AM $\perp$ BC and PN $\perp$ QR.
3. Express the ratio of areas as $\frac{\frac{1}{2} \cdot BC \cdot AM}{\frac{1}{2} \cdot QR \cdot PN}$.
4. Prove $\triangle ABM \sim \triangle PQN$ to show $\frac{AM}{PN} = \frac{AB}{PQ}$.
5. Substitute this ratio back into the area equation to get the square of the ratio.

Derivation: $\frac{ar(ABC)}{ar(PQR)} = \frac{BC \cdot AM}{QR \cdot PN} = \frac{BC}{QR} \cdot \frac{AM}{PN}$. Since $\triangle ABM \sim \triangle PQN$, $\frac{AM}{PN} = \frac{AB}{PQ}$. Also $\frac{AB}{PQ} = \frac{BC}{QR}$ (given similarity). Thus, $\frac{ar(ABC)}{ar(PQR)} = \frac{BC}{QR} \cdot \frac{BC}{QR} = \left(\frac{BC}{QR}\right)^2$. *Triangles — Areas of Similar Triangles*

10. 10. In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\triangle ABC$ is a right-angled triangle at A .

Answer:

$$\angle BAC = 90^\circ$$

Solution:

1. Observe the triangle split into two smaller triangles $\triangle ABD$ and $\triangle ACD$.
2. Write the ratio $\frac{AD}{BD} = \frac{CD}{AD}$.
3. Include the angle $\angle ADB = \angle ADC = 90^\circ$.
4. Prove $\triangle ABD \sim \triangle CAD$ by SAS similarity.
5. Equate corresponding angles $\angle BAD = \angle ACD$ and $\angle CAD = \angle ABD$.
6. Use the angle sum property of $\triangle ABC$ to show $\angle BAC = 90^\circ$.

Derivation: Given $AD^2 = BD \cdot CD \implies \frac{AD}{BD} = \frac{CD}{AD}$. In $\triangle ABD$ and $\triangle CAD$, $\angle ADB = \angle ADC = 90^\circ$. By SAS similarity, $\triangle ABD \sim \triangle CAD$. Thus $\angle BAD = \angle C$ and $\angle CAD = \angle B$. In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$. Substituting gives $\angle A + \angle CAD + \angle BAD = 180^\circ \implies 2\angle A = 180^\circ \implies \angle A = 90^\circ$. *Triangles — Similarity of Triangles*

11. 11. A vertical stick 2 m long casts a shadow 1 m long on the ground. At the same time, a tower casts a shadow 5 m long. Find the height of the tower using the properties of similar triangles.

Answer:

10meters

Solution:

1. Identify two triangles: one formed by the stick and its shadow, the other by the tower and its shadow.
2. State that at the same time, the angle of elevation of the sun is the same for both objects.
3. Identify that both triangles are right-angled and share an equal angle, thus they are similar by AA criterion.
4. Set up the proportion: $\frac{\text{Height of Stick}}{\text{Shadow of Stick}} = \frac{\text{Height of Tower}}{\text{Shadow of Tower}}$.
5. Substitute the given values and solve for the height of the tower.

Derivation: Let height of tower be h . $\triangle ABC \sim \triangle PQR$ (where ABC is stick triangle and PQR is tower triangle). Therefore, $\frac{2}{1} = \frac{h}{5}$. $h = 2 \times 5 = 10m$. *Triangles — Applications of Similar Triangles*

12. 12. The distance of the point $P(3, 4)$ from the origin is:

Answer: (C)

Solution:

1. Use the distance formula from the origin $d = \sqrt{x^2 + y^2}$
2. Substitute $x = 3$ and $y = 4$
3. Calculate $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

Derivation:

$$d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Coordinate Geometry — Distance Formula

- 13. 13.** The coordinates of the midpoint of the line segment joining the points $A(2, 4)$ and $B(4, 6)$ are:

Answer: (B)

Solution:

1. Use the midpoint formula $M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
2. Substitute the values: $x_1 = 2, x_2 = 4, y_1 = 4, y_2 = 6$
3. Calculate: $x = \frac{2+4}{2} = 3$ and $y = \frac{4+6}{2} = 5$

Derivation:

$$M = (\frac{2+4}{2}, \frac{4+6}{2}) = (3, 5)$$

Coordinate Geometry — Section Formula

- 14. 14.** The abscissa of the point $(5, 8)$ is:

Answer: (A)

Solution:

1. Recall that in a coordinate point (x, y) , x is the abscissa and y is the ordinate.
2. For the point $(5, 8)$, the x -coordinate is 5.

Derivation:

$$\text{Abscissa} = x = 5$$

Coordinate Geometry — Introduction to Cartesian Plane

- 15. 15.** If the distance between points $(2, -2)$ and $(-1, x)$ is 5, then the value of x is not asked, but what is the distance formula expression?

Answer: (D)

Solution:

1. The distance formula between two points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. Substitute $x_1 = 2, y_1 = -2, x_2 = -1, y_2 = x$

Derivation:

$$d = \sqrt{(-1 - 2)^2 + (x - (-2))^2} = \sqrt{(-3)^2 + (x + 2)^2}$$

Coordinate Geometry — Distance Formula

16. 16. The point which lies on the y-axis is:

Answer: (C)

Solution:

1. A point on the y-axis has its x -coordinate equal to 0.
2. Check the options for $x = 0$.

Derivation: Point on y-axis is of the form $(0, y)$.
Introduction to Cartesian Plane

Coordinate Geometry —

17. 17. Assertion (A): The point $(0, 5)$ lies on the y-axis. Reason (R): For any point on the y-axis, the x-coordinate is zero.

Answer: (A)

Solution:

1. Check the definition of y-axis points.
2. Verify if $(0, 5)$ satisfies the condition $x=0$.
3. Conclude that R explains A.

Derivation: A point (x, y) lies on the y-axis if and only if $x=0$. Since $(0, 5)$ has $x=0$, it lies on the y-axis.
Coordinate Geometry — Introduction to Coordinate Geometry

18. 18. Assertion (A): The distance of the point $P(3, 4)$ from the origin $(0, 0)$ is 5 units. Reason (R): The distance of a point (x, y) from the origin is calculated by $\sqrt{x^2 + y^2}$.

Answer: (A)

Solution:

1. Apply the distance formula from the origin.
2. Substitute $x=3$ and $y=4$.
3. Calculate $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Derivation:

$$d = \sqrt{(3 - 0)^2 + (4 - 0)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

Coordinate Geometry — Distance Formula

19. 19. Assertion (A): The midpoint of the line segment joining $(2, 4)$ and $(6, 8)$ is $(4, 6)$. Reason (R): The midpoint formula is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.

Answer: (A)

Solution:

1. Identify coordinates $(x_1, y_1) = (2, 4)$ and $(x_2, y_2) = (6, 8)$.
2. Use the midpoint formula.
3. Calculate $(\frac{2+6}{2}, \frac{4+8}{2}) = (4, 6)$.

Derivation:

$$M = (\frac{2+6}{2}, \frac{4+8}{2}) = (\frac{8}{2}, \frac{12}{2}) = (4, 6)$$

Coordinate Geometry — Section Formula/Midpoint

- 20. 20.** Assertion (A): The points $(1, 1)$, $(2, 2)$ and $(3, 3)$ are collinear. Reason (R): Three points are collinear if the sum of distances between any two pairs is equal to the distance between the third pair.

Answer: (B)

Solution:

1. Check the slopes: slope between $(1, 1)$ and $(2, 2)$ is $(2-1)/(2-1) = 1$.
2. Slope between $(2, 2)$ and $(3, 3)$ is $(3-2)/(3-2) = 1$.
3. Since slopes are equal, they are collinear.
4. R is a true statement, but typically collinearity is defined by slope or area = 0 in Class 10.

Derivation: Slope- $\{AB\} = 1$, Slope- $\{BC\} = 1$. Since slopes are equal, points lie on the same line.

Coordinate Geometry — Collinearity

- 21. 21.** Assertion (A): The area of a triangle with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$ is 6 sq units. Reason (R): Area of a triangle with coordinates $(0, 0)$, (x_1, y_1) , (x_2, y_2) is $\frac{1}{2}|x_1y_2 - x_2y_1|$.

Answer: (A)

Solution:

1. Use the simplified area formula for a triangle with one vertex at origin.
2. Substitute $(x_1, y_1) = (4, 0)$ and $(x_2, y_2) = (0, 3)$.
3. Calculate $\frac{1}{2}|(4)(3) - (0)(0)| = \frac{1}{2}(12) = 6$.

Derivation:

$$\text{Area} = \frac{1}{2}|4 \times 3 - 0 \times 0| = 6$$

Coordinate Geometry — Area of Triangle

- 22. 22.** Find the ratio in which the line segment joining the points $A(-3, 10)$ and $B(6, -8)$ is divided by the point $P(-1, 6)$.

Answer:

2 : 7

Solution:

1. Use the section formula: $P(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
2. Substitute the coordinates and assume ratio $k : 1$
3. Solve for k using the x-coordinate equation
4. Verify with the y-coordinate

Derivation:

$$\begin{aligned}
 -1 &= \frac{k(6) + 1(-3)}{k + 1} \\
 \implies -k - 1 &= 6k - 3 \\
 \implies 2 &= 7k \\
 \implies k &= \frac{2}{7}
 \end{aligned}$$

Coordinate Geometry — Section Formula

- 23. 23.** If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p .

Answer:

$$p = 7$$

Solution:

1. Recall that diagonals of a parallelogram bisect each other.
2. Find the midpoint of diagonal AC.
3. Find the midpoint of diagonal BD.
4. Equate the midpoints and solve for p .

Derivation: Midpoint of AC = $\left(\frac{6+9}{2}, \frac{1+4}{2} \right) = (7.5, 2.5)$; Midpoint of BD = $\left(\frac{8+p}{2}, \frac{2+3}{2} \right) = \left(\frac{8+p}{2}, 2.5 \right)$; $\frac{8+p}{2} = 7.5 \implies 8+p = 15 \implies p = 7$ *Coordinate Geometry — Midpoint Formula*

- 24. 24.** Find the distance between the points $P(a + b, b - a)$ and $Q(a - b, a + b)$.

Answer:

$$2\sqrt{a^2 + b^2}$$

Solution:

1. Apply the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. Substitute the given coordinates.
3. Simplify the algebraic terms inside the square root.
4. Calculate the square root.

Derivation:

$$d = \sqrt{((a-b)-(a+b))^2 + ((a+b)-(b-a))^2} = \sqrt{(-2b)^2 + (2a)^2} = \sqrt{4b^2 + 4a^2} = 2\sqrt{a^2 + b^2}$$

Coordinate Geometry — Distance Formula

- 25. 25.** Find the value of y for which the distance between the points $P(2, -3)$ and $Q(10, y)$ is 10 units.

Answer:

$$y = 3 \text{ or } y = -9$$

Solution:

1. Use the distance formula $PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
2. Substitute the known values into the equation.
3. Solve the resulting quadratic equation for y .

Derivation:

$$\begin{aligned} 10^2 &= (10 - 2)^2 + (y - (-3))^2 \\ \implies 100 &= 64 + (y + 3)^2 \\ \implies 36 &= (y + 3)^2 \\ \implies y + 3 &= \pm 6 \\ \implies y &= 3, -9 \end{aligned}$$

Coordinate Geometry — Distance Formula

- 26. 26.** Find the coordinates of a point A , where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.

Answer:

$$A(3, -10)$$

Solution:

1. Let A be (x, y) . The centre $(2, -3)$ is the midpoint of diameter AB .
2. Use the midpoint formula: $\frac{x+1}{2} = 2$ and $\frac{y+4}{2} = -3$.
3. Solve for x and y .

Derivation:

$$\begin{aligned} x + 1 &= 4 \\ \implies x &= 3; y + 4 = -6 \\ \implies y &= -10. \text{ Point A is } (3, -10) \end{aligned}$$

Coordinate Geometry — Midpoint Formula

- 27. 27.** Find the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$. Also, find the value of x .

Answer:

$$\text{Ratio is } 3 : 5, x = 9$$

Solution:

1. Use the section formula for the y-coordinate to find the ratio $k:1$.
2. Set the y-coordinate formula equal to 2.
3. Solve for k .
4. Use the value of k to find the x-coordinate using the section formula.
5. State the final ratio and the value of x .

Derivation: Let the ratio be $k:1$. The y-coordinate is given by $y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$.
 Substituting values: $2 = \frac{k(-3) + 1(5)}{k + 1} \implies 2k + 2 = -3k + 5 \implies 5k = 3 \implies k = 3/5$.
 Thus, the ratio is $3:5$. For x : $x = \frac{3(4) + 5(12)}{3 + 5} = \frac{12 + 60}{8} = \frac{72}{8} = 9$. *Coordinate Geometry — Section Formula*

- 28. 28.** If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p .

Answer:

$$p = 7$$

Solution:

1. Recall the property that diagonals of a parallelogram bisect each other.
2. Find the midpoint of diagonal AC.
3. Find the midpoint of diagonal BD in terms of p .
4. Equate the midpoints and solve for p .

Derivation: Midpoint of AC = $(\frac{6+9}{2}, \frac{1+4}{2}) = (7.5, 2.5)$. Midpoint of BD = $(\frac{8+p}{2}, \frac{2+3}{2}) = (\frac{8+p}{2}, 2.5)$. Equating x-coordinates: $7.5 = \frac{8+p}{2} \implies 15 = 8 + p \implies p = 7$. *Coordinate Geometry — Midpoint Formula*

- 29. 29.** Find the area of a quadrilateral whose vertices, taken in order, are $(-4, -2)$, $(-3, -5)$, $(3, -2)$, and $(2, 3)$.

Answer:

$$28 \text{ sq. units}$$

Solution:

1. Divide the quadrilateral into two triangles by drawing a diagonal.
2. Use the area of triangle formula: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
3. Calculate the area of triangle 1.

4. Calculate the area of triangle 2.
5. Add the two areas to get the total area.

Derivation: Area of $\triangle ABC$ with vertices $(-4, -2), (-3, -5), (3, -2)$: $\frac{1}{2}|-4(-5 - (-2)) + (-3)(-2 - (-2)) + 3(-2 - (-5))| = \frac{1}{2}|12 + 0 + 9| = 10.5$. Area of $\triangle ADC$ with vertices $(-4, -2), (2, 3), (3, -2)$: $\frac{1}{2}|-4(3 - (-2)) + 2(-2 - (-2)) + 3(-2 - 3)| = \frac{1}{2}|-20 + 0 - 15| = 17.5$. Total Area = $10.5 + 17.5 = 28$. *Coordinate Geometry — Area of Triangle*

- 30. 30.** Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

Answer:

$$(-1, 3.5), (0, 5), (1, 6.5)$$

Solution:

1. Identify the midpoints needed to divide the segment into 4 parts.
2. Find the midpoint M of AB.
3. Find the midpoint of AM to get the first point.
4. Find the midpoint of MB to get the third point.
5. List all three points.

Derivation: Midpoint M of AB = $(\frac{-2+2}{2}, \frac{2+8}{2}) = (0, 5)$. Midpoint P of AM = $(\frac{-2+0}{2}, \frac{2+5}{2}) = (-1, 3.5)$. Midpoint Q of MB = $(\frac{0+2}{2}, \frac{5+8}{2}) = (1, 6.5)$. The points are $(-1, 3.5), (0, 5), (1, 6.5)$. *Coordinate Geometry — Section Formula Application*

- 31. 31.** If $\sin \theta = \frac{3}{5}$, then the value of $\cos \theta$ is:

Answer: (C)

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$
2. Substitute the value of $\sin \theta$
3. Solve for $\cos \theta$

Derivation:

$$\begin{aligned}\cos^2 \theta &= 1 - \sin^2 \theta = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25} \\ \implies \cos \theta &= \frac{4}{5}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

- 32. 32.** The value of $\tan 45^\circ + \cot 45^\circ$ is:

Answer: (B)

Solution:

1. Recall the trigonometric values for standard angles.
2. Substitute $\tan 45^\circ = 1$ and $\cot 45^\circ = 1$.
3. Add the values.

Derivation:

$$\tan 45^\circ + \cot 45^\circ = 1 + 1 = 2$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 33. 33.** If $3 \tan \theta = 3$, then θ is equal to:

Answer: (C)

Solution:

1. Simplify the equation to find $\tan \theta$.
2. Identify the angle θ for which $\tan \theta = 1$.

Derivation:

$$\begin{aligned} 3 \tan \theta &= 3 \\ \implies \tan \theta &= 1 \\ \implies \theta &= 45^\circ \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

- 34. 34.** The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is:

Answer: (A)

Solution:

1. Use the complementary angle identity $\cos(90^\circ - \theta) = \sin \theta$.
2. Substitute $\cos 72^\circ = \cos(90^\circ - 18^\circ) = \sin 18^\circ$.
3. Divide the terms.

Derivation:

$$\frac{\sin 18^\circ}{\cos 72^\circ} = \frac{\sin 18^\circ}{\sin 18^\circ} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

- 35. 35.** Which of the following is equal to $\sec^2 \theta - 1$?

Answer: (D)

Solution:

1. Recall the fundamental trigonometric identity $1 + \tan^2 \theta = \sec^2 \theta$.
2. Rearrange the identity to isolate $\sec^2 \theta - 1$.

Derivation:

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\implies \sec^2 \theta - 1 = \tan^2 \theta$$

Introduction to Trigonometry — Trigonometric Identities

- 36. 36.** Assertion (A): If $\sin \theta = \frac{1}{2}$, then $\theta = 30^\circ$. Reason (R): The value of $\sin \theta$ increases as θ increases from 0° to 90° .

Answer: (B)

Solution:

1. Check the validity of Assertion: For $\sin \theta = 1/2$, $\theta = 30^\circ$ is true.
2. Check the validity of Reason: The sine function is strictly increasing in the first quadrant, which is true.
3. Determine if R explains A: The fact that $\sin 30^\circ = 1/2$ is a specific value, its increasing nature is a general property, so R does not explain why $\sin 30^\circ$ is specifically $1/2$.

Derivation:

$$\text{Assertion: } \sin 30^\circ = 0.5$$

$$\implies \text{True. Reason: } \frac{d}{d\theta}(\sin \theta) = \cos \theta > 0 \text{ for } \theta \in (0, 90^\circ)$$

$$\implies \text{True.}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 37. 37.** Assertion (A): The value of $\cos \theta$ can never be 1.5. Reason (R): The range of $\cos \theta$ is $[-1, 1]$.

Answer: (A)

Solution:

1. Check the validity of Assertion: Since the range of cosine is $[-1, 1]$, 1.5 is outside this range, so it is true.
2. Check the validity of Reason: By definition of trigonometric ratios in a right triangle, $\cos \theta = \text{base/hypotenuse} \leq 1$.
3. Determine if R explains A: The range definition is the direct reason why $\cos \theta$ cannot exceed 1.

Derivation:

$$-1 \leq \cos \theta \leq 1 \text{ for all } \theta \in \mathbb{R}$$

Introduction to Trigonometry — Basic Trigonometric Ratios

- 38. 38.** Assertion (A): $\sin^2 \theta + \cos^2 \theta = 1$ for all θ . Reason (R): $\tan^2 \theta - \sec^2 \theta = 1$.

Answer: (C)

Solution:

1. Check the validity of Assertion: This is the fundamental Pythagorean identity, which is true.
2. Check the validity of Reason: The identity is $\sec^2 \theta - \tan^2 \theta = 1$. Therefore, $\tan^2 \theta - \sec^2 \theta = -1$. The reason is false.

Derivation:

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \text{ (Identity); } \sec^2 \theta - \tan^2 \theta = 1 \\ &\implies \tan^2 \theta - \sec^2 \theta = -1\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

39. 39. Assertion (A): If $\tan A = \cot A$, then $A = 45^\circ$. Reason (R): $\tan A = \cot(90^\circ - A)$.

Answer: (A)

Solution:

1. Check the validity of Assertion: If $\tan A = \cot A$, then $\tan A = 1/\tan A$, so $\tan^2 A = 1$, $\tan A = 1$, $A = 45^\circ$. True.
2. Check the validity of Reason: This is the standard co-function identity.
3. Determine if R explains A: Yes, $\tan A = \cot A \implies \tan A = \tan(90^\circ - A) \implies A = 90^\circ - A \implies 2A = 90^\circ \implies A = 45^\circ$.

Derivation:

$$\begin{aligned}\tan A &= \cot A \\ \iff \tan A &= \tan(90^\circ - A) \\ \iff A &= 90^\circ - A \\ \iff A &= 45^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

40. 40. Assertion (A): The value of $\sin 0^\circ$ is 0. Reason (R): The value of $\cos 0^\circ$ is 0.

Answer: (C)

Solution:

1. Check the validity of Assertion: $\sin 0^\circ = 0$. True.
2. Check the validity of Reason: $\cos 0^\circ = 1$. False.

Derivation:

$$\sin 0^\circ = 0, \cos 0^\circ = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles